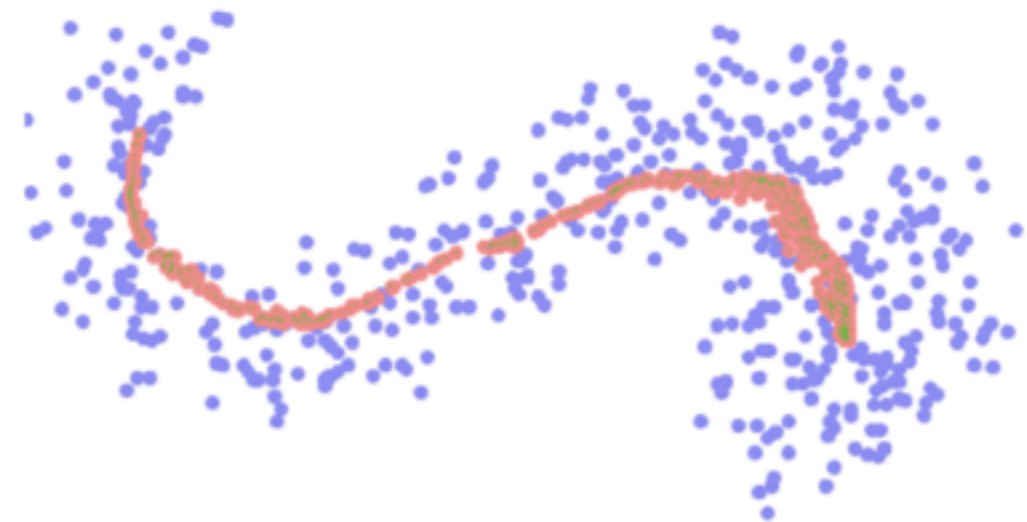


Interpolates

Interpolates

$$\min_{\alpha} W_{\varepsilon,p}^p(\alpha, \beta)$$



$$\min_{\alpha} \overline{W}_{\varepsilon,p}^p(\alpha, \beta)$$



α

β

Sinkhorn Divergence

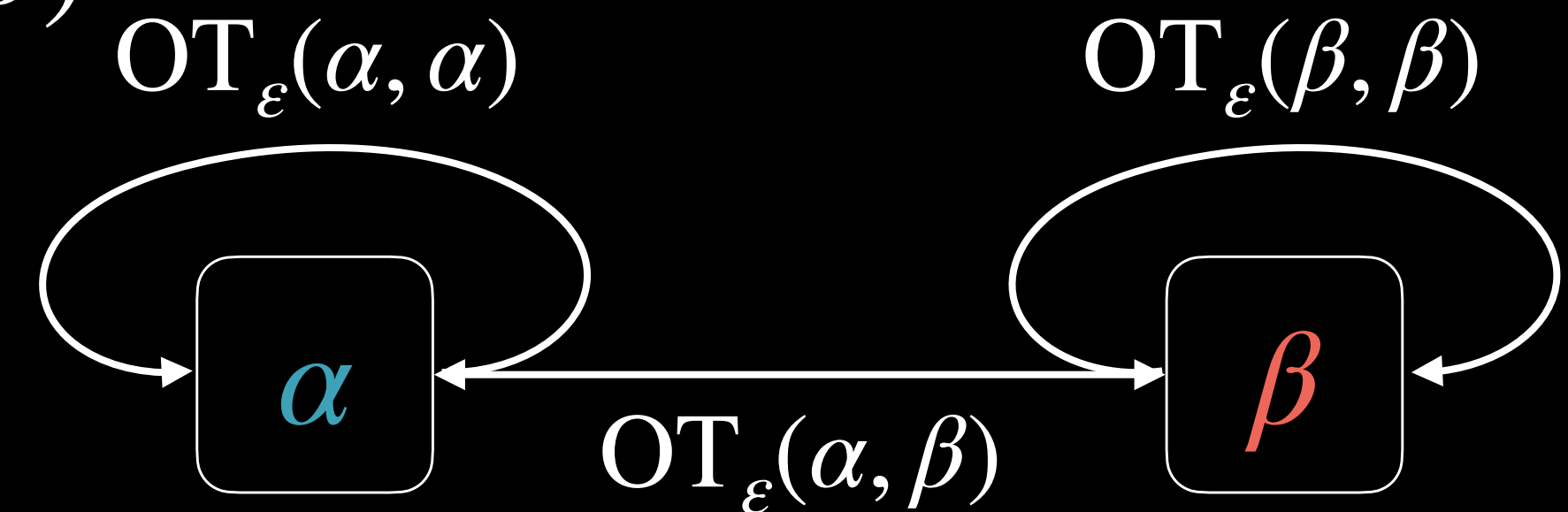
- **Sinkhorn Divergence** [Raudas et al. 2017; Salimans et al. 2018; Genevay et al. 2018]:

$$S_\varepsilon(\alpha, \beta) = \text{OT}_\varepsilon(\alpha, \beta) - \frac{1}{2}\text{OT}_\varepsilon(\alpha, \alpha) - \frac{1}{2}\text{OT}_\varepsilon(\beta, \beta)$$

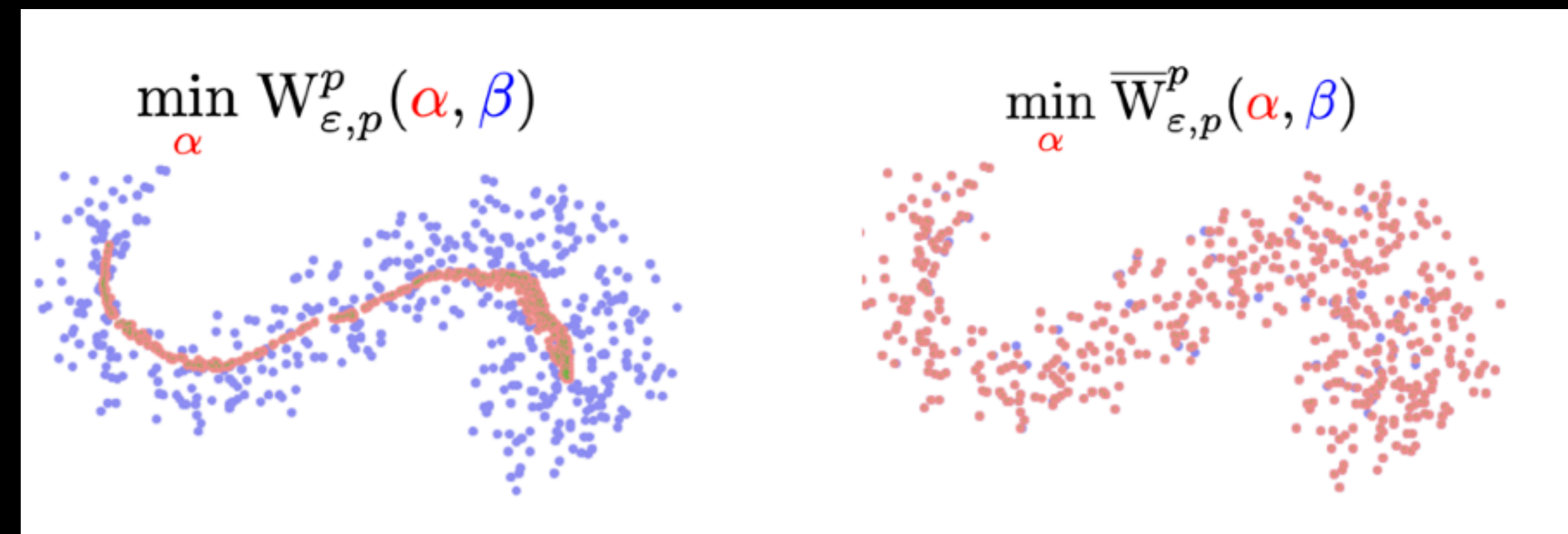
- Properties:

- Symmetric, nonnegative, positive definite.

- **Interpolates** between: $\text{MMD} (\varepsilon \rightarrow \infty) \leftrightarrow \text{OT} (\varepsilon \rightarrow 0)$.



- Metric-like, stable in ML practice.



Toy Example: 1D Gaussians

$$\alpha = \mathcal{N}(0, 1) \quad \beta = \mathcal{N}(\Delta, 1)$$

